

ANALYSTLAB AFRICA EXPERIENCE LAB

Official Week 5 Project Summary — Machine Learning Engineering Track

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Track: Machine Learning Engineering / Data Science

Project Title: HealthConnect Clinic Appointment Attendance Optimization

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1. What I Planned to Accomplish in Week 5

Building on the Week 4 problem definition and production system design, I planned to move from planning into implementation by: constructing modular, reusable data-processing and preprocessing pipelines; engineering behavioural features; training and benchmarking baseline classifiers on recall-focused metrics; evaluating initial results; and testing the individual pipeline components.

2. What I Actually Completed

1. A modular **ML pipeline package** (pipeline/) with single-responsibility modules for configuration, data processing, feature engineering, model training and evaluation.
2. A **data-processing workflow** with explicit leakage control (dropping `waiting_time_minutes`), NMAR/MCAR imputation, and a reusable sklearn `ColumnTransformer`.
3. **Feature engineering** adding four behavioural features.
4. An **initial model workflow** training and benchmarking **five** baselines on 5-fold stratified cross-validation.
5. An **evaluation framework** with metrics (accuracy, precision, recall, F1, ROC-AUC) and diagnostic visualisations.
6. **Component testing** — all 6 smoke tests pass.
7. Documents: `README.md`, `requirements.txt`, this report and a project summary (LaTeX/PDF), plus a fully executed Jupyter notebook.

3. Key Findings / Development Outcomes

- The primary baseline (Logistic Regression) achieves held-out **recall** 0.650, **accuracy** 0.617, and **ROC-AUC** 0.664, comfortably above the random baseline and confirming pre-appointment signal.
- `booking_lead_days` dominates feature importance, followed by `distance_to_clinic_km`, `age`, and appointment history — consistent with Week 4 findings.
- All five baselines are broadly comparable in recall (0.56–0.63), indicating a reliable starting point for tuning rather than a final model.

4. Major Challenges Encountered

- Designing preprocessing and imputation to preserve the meaning of NMAR missingness.
- Managing the dataset reduction (5,000 $\rightarrow$ 4,737 rows) after dropping cancellations.
- Handling the `booking_lead_days = 0` binning edge case to avoid NaN labels.
- Acknowledging that a stratified split does not fully control repeat-patient leakage (needs `GroupKFold`).

5. Important Decisions Made and Why

- **Selected logistic regression as the primary reporting baseline** for interpretability at this initial stage.
- **Used stratified split for the baseline** to establish performance quickly, with `GroupKFold` explicitly deferred to Week 6.
- **Used balanced class weighting** to favour no-show recall over precision, matching the business objective of catching high-risk patients.

6. Changes to My Week 4 Approach

The Week 4 architecture and problem definition were followed unchanged. The main refinement was choosing an interpretable linear baseline for the initial implementation while retaining tree-based models for later feature-importance and hyperparameter work.

7. Cross-Track Collaboration

Liaised with the Data Science / Data Analytics tracks to confirm the target definition (Attended vs No-Show), the exclusion of cancellations, and the leakage boundaries. This ensured the engineered features and pipeline align with the analytics outputs produced across the project team.

8. Remaining Work

Notes: apply `GroupKFold` and a temporal split; tune hyperparameters and threshold for the F_2 objective; containerise the inference service; add MLflow tracking and drift monitoring.

9. Proposed Focus for Week 6

1. Rigorous leakage-controlled validation (`GroupKFold` by `patient_id`).
2. Hyperparameter and threshold tuning for recall-driven objectives.
3. Wrapping the best model in the containerised FastAPI inference service from the Week 4 architecture.
4. Introducing MLflow experiment tracking and Evidently AI drift monitoring.